

PITE-RONNSKAR, Sweden

WMO: 021760

Lat: 65.034N

Lon: 21.561E

Elev: 16

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.9	-4.7	-17.7	2.1	-5.6	-10.1	3.2	-4.1	35.4	28.2	32.1	22.9	5.9	220	0.770

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	70.4	63.7	67.6	61.4	65.5	60.0	65.4	69.0	63.2	66.4	61.4	64.4	8.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	88.8	67.9	61.7	82.3	65.4	59.9	77.0	63.4	30.3	68.9	28.7	66.4	27.4	64.5	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.7	27.2	24.0	DB	-15.9	75.1	10.7	3.5	-23.5	77.6	-29.8	79.7	-35.8	81.7	-43.6	84.2	(4)
(5)			WB	-15.9	66.2	9.9	2.6	-23.1	68.1	-28.9	69.6	-34.5	71.1	-41.7	72.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.1	19.8	17.2	24.5	33.3	43.4	53.2	60.1	57.9	49.9	39.8	31.1	25.0	(6)
(7)		DBStd	16.29	11.78	12.65	8.92	5.43	5.80	4.94	4.38	4.37	4.93	6.17	7.79	10.07	(7)
(8)		HDD50	5106	935	918	791	500	217	23	0	2	62	318	566	774	(8)
(9)		HDD65	9846	1400	1338	1256	950	668	355	165	224	452	783	1016	1239	(9)
(10)		CDD50	751	0	0	0	0	14	118	312	247	59	1	0	0	(10)
(11)		CDD65	16	0	0	0	0	0	0	12	4	0	0	0	0	(11)
(12)		CDH74	25	0	0	0	0	2	0	18	5	0	0	0	0	(12)
(13)		CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0	(13)
(14)	Wind	WSAvg	11.2	12.2	11.2	11.2	9.9	10.1	10.0	9.9	10.6	11.2	12.2	12.7	12.9	(14)
(15)	Precipitation	PrecAvg	20.0	1.3	0.9	0.8	1.1	1.3	2.0	2.5	2.3	2.0	2.0	1.5	(15)	
(16)		PrecMax	26.7	2.8	1.9	2.0	2.2	3.5	3.5	6.3	4.3	6.2	6.1	3.8	3.9	(16)
(17)		PrecMin	13.5	0.2	0.2	0.2	0.2	0.2	0.4	0.9	0.4	0.4	0.5	0.5	(17)	
(18)		PrecStd	3.8	0.6	0.4	0.5	0.5	0.8	0.8	1.2	1.2	1.3	1.2	0.9	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	44.4	47.4	53.7	64.9	68.3	75.6	72.0	63.9	54.4	47.9	43.0	(19)
(20)			MCWB	38.4	39.1	39.7	43.0	51.8	56.3	67.2	65.1	54.1	47.8	43.4	39.3	(20)
(21)		2%	DB	38.0	39.5	43.1	48.0	58.2	65.1	72.1	68.7	60.4	51.1	44.1	40.2	(21)
(22)			MCWB	34.7	35.3	36.8	39.8	48.6	55.0	65.6	63.0	54.4	48.0	42.2	37.7	(22)
(23)		5%	DB	35.1	35.3	39.4	43.6	54.3	62.6	69.3	66.4	58.5	49.6	42.6	38.3	(23)
(24)			MCWB	33.3	32.7	34.3	37.8	46.9	54.9	63.2	61.2	54.7	47.6	41.2	36.3	(24)
(25)		10%	DB	33.3	32.7	35.5	40.1	51.2	60.5	66.7	64.6	57.0	48.1	40.8	36.7	(25)
(26)			MCWB	32.0	31.2	32.5	36.1	45.4	54.3	61.6	60.1	54.0	46.0	39.3	35.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.6	39.4	40.6	43.9	53.9	60.8	69.6	66.6	58.5	50.6	44.1	40.1	(27)
(28)			MCD B	42.5	43.9	46.7	51.2	61.0	63.9	73.4	70.4	59.7	51.3	45.5	41.6	(28)
(29)		2%	WB	35.2	35.7	37.4	40.9	50.4	58.3	67.0	64.4	56.8	49.3	42.9	38.3	(29)
(30)			MCD B	36.9	39.2	42.1	46.3	56.3	61.9	70.9	67.6	58.1	50.3	44.1	39.8	(30)
(31)		5%	WB	33.7	33.1	34.8	38.5	48.0	56.9	64.5	62.6	55.9	47.9	41.4	36.8	(31)
(32)			MCD B	35.0	34.7	38.1	42.5	53.1	60.6	67.9	65.4	57.4	49.5	42.4	38.0	(32)
(33)		10%	WB	32.2	31.5	33.1	36.6	45.9	55.2	62.5	61.2	54.7	46.0	39.7	35.4	(33)
(34)			MCD B	33.2	32.8	35.1	39.6	50.4	59.0	65.9	63.7	56.5	47.8	40.7	36.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.1	11.7	11.9	9.4	8.7	7.8	7.5	7.8	8.4	7.4	6.9	8.4	(35)
(36)			MCDBR	8.9	12.5	15.8	15.5	15.2	11.6	10.8	10.0	10.3	8.2	6.6	7.1	(36)
(37)			MCWBR	8.0	10.4	11.3	9.5	8.6	6.4	6.8	7.0	6.9	6.4	5.8	6.5	(37)
(38)		5% WB	MCDBR	8.7	11.8	13.8	14.2	13.5	8.2	8.4	7.7	7.3	6.5	6.1	6.7	(38)
(39)			MCWBR	8.0	10.1	10.4	9.2	8.3	5.7	6.3	6.0	5.9	5.7	5.7	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub		N/A	0.275	0.303	0.337	0.344	0.348	0.363	0.350	0.321	0.302	0.339	0.348	(40)
(41)		taud		N/A	2.145	2.204	2.274	2.449	2.495	2.472	2.501	2.556	2.389	2.598	2.620	(41)
(42)		Ebn at Noon		N/A	193	243	263	274	275	266	259	244	188	91	38	(42)
(43)		Edh at Noon		N/A	18	27	32	30	30	30	26	20	15	7	4	(43)
(44)	All-Sky Solar Radiation	RadAvg		36	219	698	1239	1570	1735	1603	1199	692	265	58	10	(44)
(45)		RadStd		3	37	66	94	157	139	149	111	92	47	10	1	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page